

Make Your Own Smart Light

A. Samiuddhin

Most textile retail stores have trial/changing rooms for customers to try on new clothes before buying them. These rooms are illuminated all day long, irrespective of whether these are being used or not. This leads to wastage of electricity.

Presented here is the circuit that turns the light on only when someone enters the room. This circuit also indicates the occupancy status of the room. It can be used for bathrooms and public toilets as well, besides some other places.

Circuit and working

The circuit diagram of the smart light is shown in Fig. 1. It is built around 5V voltage regulator 7805 (IC1), AVR microcontroller (MCU) ATtiny85A

(IC2), opto-coupler PC817 (IC3), PIR sensor, common-cathode bi-colour LED (BICO1), normally-open magnetic reed switch (S2), 12V single-changeover relay (RL1), two BC547 npn transistors (T1 and T2) and a few other components.

IC2 is the brain of the circuit, which turns on/off the light connected through the relay based on input state. Presence of a person in the room is detected by the PIR sensor, whereas door status is detected by the reed switch. If any motion is detected by the sensor, its output goes high. Otherwise, the output remains low.

Reed switch contacts remain open when the door is open. When door is closed, contacts get closed. Statuses of the PIR sensor, reed switch and relay

are given in the table. It gives input and corresponding output statuses for every action.

If the room is unoccupied and the door is closed, the LED emits green light. The relay switches off to save power. If someone opens the door, the LED emits red and green colours alternately until the person enters the room.

The relay turns on for about ten seconds and then it turns off. Light bulb (Load1) connected through the relay illuminates during this period. If a person enters the room within ten seconds after the door is opened, the relay gets activated and LED emits orange colour until the door is closed. Thereafter the LED emits red colour, indicating that the room is occupied and the door is closed.

The MCU is isolated from the relay power supply using opto-coupler IC3. The relay is turned on/off through the opto-coupler. This protects the MCU from any disturbance in the 12V supply for the relay. Freewheeling diode D1 suppresses the induced high voltage produced by the relay coil when relay RL1 is turned off. In the circuit, Load1 is the main light, while Load2 is the optional standby light.

Status of PIR Sensor, Reed Switch and Bi-Colour LED				
Actions	Inputs		Outputs	
	PIR sensor	Reed switch	Bi-colour LED (BICO1)	Relay
Room empty	Idle	Closed	Green colour glows	Off
Door open	Idle	Opened	Red and green colours blink alternately	On for about ten seconds
Person enters/exits	Detects motion	Opened	Orange colour glows	On
Room occupied	Detects motion	Closed	Red colour glows	On

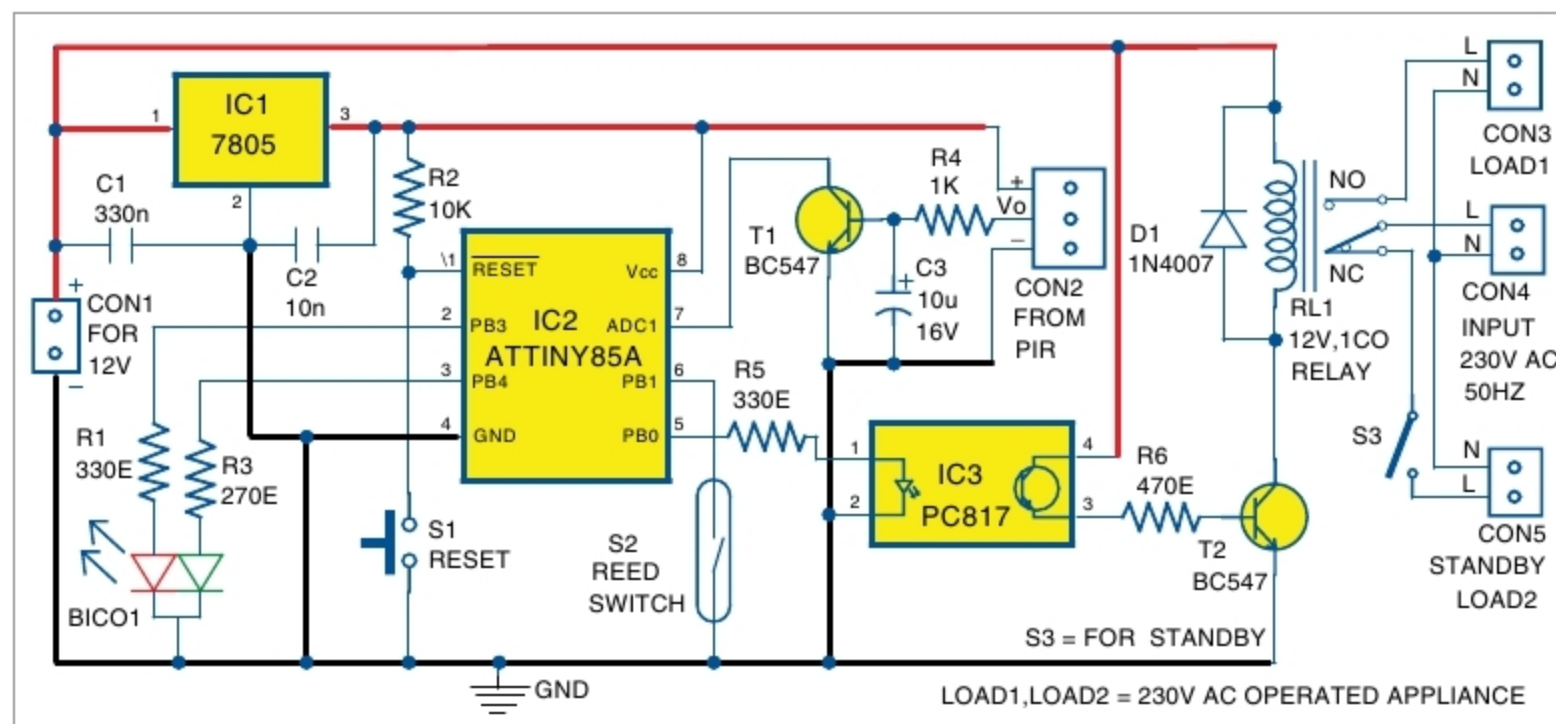


Fig. 1: Circuit diagram of the smart light

Software program

The source code (trial_room.ino) is developed using Arduino IDE. By default, Arduino IDE does not support ATtiny85A and, hence, we need to add ATtiny library to Arduino IDE.

Adding ATtiny support. To add ATtiny support to Arduino IDE, follow the